Beyond Prediction: Prescriptive AI for Energy Portfolio Optimization

An AI-assisted decision system for risk-aware capital allocation across oil, gas, power, renewables and storage

Alan Mössinger · SIADS 699 Capstone, University of Michigan · Group 13 (a solo project) · 2026

The decision

One budget, forty candidates, 1 January 2025, and only what was public that day: twenty Gulf of Mexico lease redevelopments and twenty ERCOT interconnection requests still waiting in the queue. A committee must choose how much expected value to give up for a thinner tail, and how to fund the construction its requests may call. Four questions: can machine learning **predict** better than transparent baselines, how does risk change value, which book should be **prescribed**, and how much does each part — the forecasts, the optimizer, the language model — add?

Prediction is not the decision

Leases add value; wind and solar requests pay only under contract

Left: incremental NPV against the Arps baseline, base case with the high-cost to low-cost span (costs ASSUMED, production FORECAST). Right: expected NPV on 1 January 2025 = P(operation by year, given the age in the queue) × NPV if built − cost to carry; 2024 ERCOT North basis, ATB costs in 2024 dollars, credits as earned.

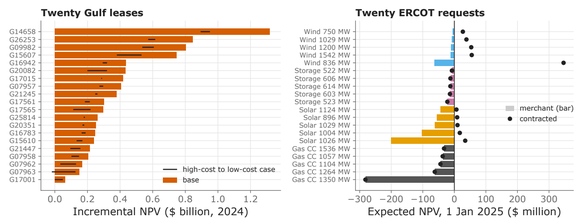


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: ASSUMED. Cohort: the default book: the 20 leases with the most output in the twelve months published by 1 January 2025, and the five largest ERCOT requests of each technology still in the queue that day

Every lease adds value at the base cost; wind and solar requests pay only under contract. Backtested as of 2020, the queue model ranked requests better than a lookup (AUC 0.731 against 0.686) — yet the book chosen with its curves scores +$22M [−$168M, +$207M] against the book chosen with the lookup's: not demonstrably better. The decision turns on a few large requests whose chance of operating neither method knows well.

What each part adds, on the same scenarios

The risk-aware book trades expected value for a better risk-adjusted score

Each comparison on the same 2,000 held-out scenarios with paired bootstrap intervals; committee grid, contracted case. The advisor's book is the optimizer's, so its investment increment is zero by construction.

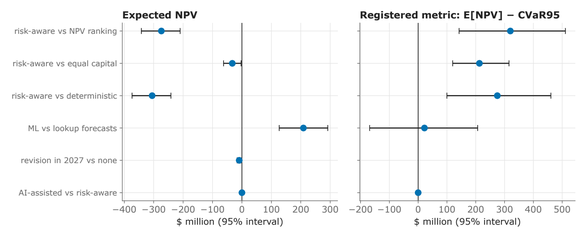


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025

The optimizer adds value only when held to the same constraints as the rules it is compared with. Revising the book in 2027 is worth −$9M; knowing everything in advance would be worth at most $571M.

The system in one picture

From prediction to a challenged, human, measured decision

The decision system: each stage is a tested module of src/; arrows show what each hands the next.

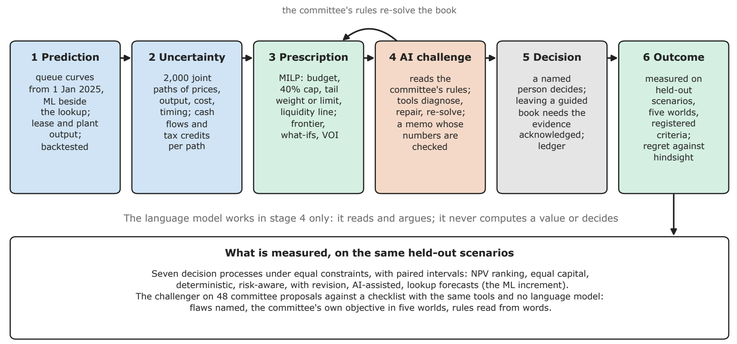


Figure: Alan Mössinger. Data: LBNL Queued Up: 2025 Edition (Rand et al., 2025). Provenance: DERIVED. Cohort: the forty candidates on 1 January 2025

The language model works in one stage only: it reads and argues; it never computes a value or decides. A named person signs, and the ledger records every decision with what it saw.

The trade-off the committee must price

Under one cap, risk aversion trades expected value for a thinner tail

Left: books chosen on 2,000 primary scenarios, scored on 2,000 held-out ones; committee grid, contracted case; budget $3.42B, 40% of the forty stage-one charges; no technology above 40% of it. Right: what each distinct book on the λ path spends, by technology.

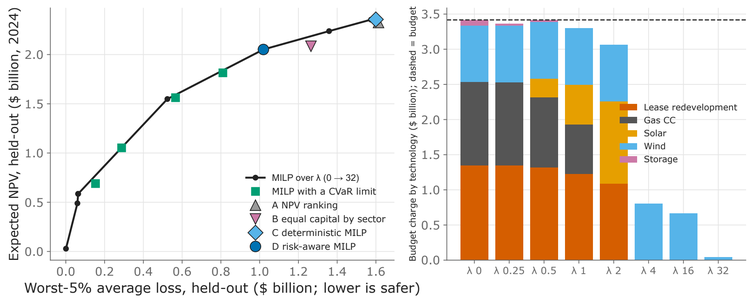


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: the forty candidates on 1 January 2025 (notebook 07); scenarios of notebook 08

On 2,000 held-out scenarios the risk-aware book expects $2.05B with a worst-5% loss of $1.02B; the return-seeking book $2.36B with $1.60B. Under equal constraints the risk-aware book beats NPV ranking by $320M [+$142M, +$511M] on expected NPV minus the tail, and does so in all 20 re-selections.

Does an AI challenger improve the committee's decision?

Challenging adds $192M per case; tools alone add $145M

Left: share of the four cases of each flaw type in which the memo (blue) or the checklist control (grey) names the flaw. Right, above the dotted line: the committee's stated objective (its tail weight, funded within its liquidity line) for the book it ends with, per flawed case, mean of five worlds; *a fixed rule that takes the optimizer's book first. Below: the worst-5% loss avoided in the base world. 36 flawed cases. Simulated committees (ten personas); post-registration.

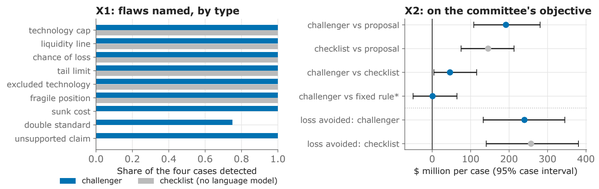


Figure: Alan Mössinger. Data: BSEE OGOR-A production (data.bsee.gov). ERCOT DAM/RTM settlement point prices (gridstatus pull, 2024; ERCOT yearly DAM files, 2018–2023). Provenance: FORECAST. Cohort: 48 committee cases on the forty candidates of 1 January 2025

Registered before it ran, on 48 proposals from simulated committees. Following the challenger improved the committee's own objective by $192M per flawed case [+$108M, +$281M] and avoided $240M of worst-5% loss. A checklist with the same tools and no language model achieved $145M and $257M: most of the gain is the tools'. What only the language model adds is understanding — 11 of 12 flaws of argument named (the checklist 0), 40 of 40 rules read from words. It over-challenged 2 of 12 sound proposals; 6 of 8 registered lines pass.

The capital plan is the weak point

The reserve is exceeded in 44% of scenarios; 11 of book D's 13 members are core

Left: book D's construction calls beyond its reserve, held-out scenarios. Right: how often each candidate is chosen when book D is re-chosen on 20 resampled primary sets; dotted line = core (80%). Committee grid, contracted case.

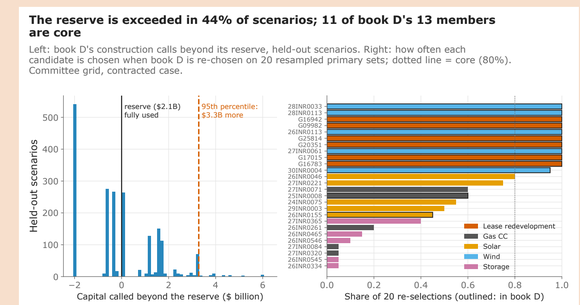


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Construction calls exceed the reserve in 44.5% of scenarios; without a contingent line the requests that reach their agreement cannot be built and the book is worth $1.40B.

Recommendation

Choose the tail weight first. At the registered weight, fund the risk-aware book — six leases, five wind, one solar and one gas request — with a contingent construction line of $3.28B, which restores $2.02B of value. Defer the requests whose agreement signal is worth more than their deposit. **Use the challenger before signing**, and record any departure from its guidance with the evidence acknowledged.

Prediction is necessary and not sufficient: the decision is made by the constraints, the capital plan and the tail — and a committee is helped most by tools that test its own rules. Model-based results on public data and assumed lease costs; simulated committees. Data: LBNL Queued Up 2025 (Rand et al., 2025); BSEE OGOR-A; EIA; ERCOT settlement-point prices; NREL ATB. Author's analysis.